FA4

Joshua Hermosada, https://github.com/diousoix/Probability-Probability-Distribution-FAs.git

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Item #2

```
zero <- 0.70
one <- 1 - zero

received_zero_given_zero <- 0.95
received_one_given_one <- 0.75

given_zero_received_one <- 1 - received_zero_given_zero
given_one_received_zero <- 1 - received_one_given_one

one_is_received <- (zero * given_zero_received_one) + (one * received_one_given_one)
one_is_received
```

```
## [1] 0.26
```

```
transmitted_one_received_one <- (one * received_one_given_one) / one_is_received
transmitted_one_received_one
```

```
## [1] 0.8653846
```

Item #7

```
jane <- 0.10
amy <- 0.30
ava <- 0.60

jane_error <- 0.08
amy_error <- 0.05
ava_error <- 0.01

everybody_error <- (jane * jane_error) + (amy * amy_error) + (ava * ava_error)
everybody_error
```

```
## [1] 0.029
```

```
is_it_jane <- (jane * jane_error) / everybody_error
is_it_jane
```

```
## [1] 0.2758621
```

```r
is_it_amy <- (amy * amy_error) / everybody_error
is_it_amy
```

```
## [1] 0.5172414
```

```r
is_it_ava <- (ava * ava_error) / everybody_error
is_it_ava
```

```
## [1] 0.2068966
```

After a thorough investigation, it has been determined that Amy would most likely be the person to have written the program error with a probability of 51.72%